

ANALYSTLAB AFRICA
EXPERIENCE LAB INTERNSHIP PROGRAMME

HEALTHCONNECT CLINIC EXPERIENCE LAB

INITIAL RISK & DEPENDENCY REGISTER

Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Document Control

Project	HealthConnect Clinic Experience Lab
Programme	AnalystLab Africa Experience Lab
Document Type	Initial Risk & Dependency Register
Version	1.0
Project Phase	Initiation / Planning
Issue Date	August 2026
Status	Initial Baseline

1. RISK REGISTER

ID	Risk	Cause / Trigger	Potential Impact	Likelihood	Impact	Rating	Response Strategy	Response / Mitigation	Owner	Status
R-01	Insufficient or poor-quality data	Missing, inconsistent or incomplete appointment records	Weak analysis and unreliable predictions	High	High	Critical	Mitigate	Conduct early data profiling; document gaps; define minimum data requirements	Data & Analytics Lead	Open
R-02	Limited visibility into actual clinic workflows	Insufficient access to operational processes or users	Solution may not fit real operating conditions	Medium	High	High	Mitigate	Conduct workflow discovery and validate assumptions with operational stakeholders	Project Manager	Open
R-03	Predictive model produces unreliable classifications	Weak features, class imbalance or insufficient training data	Incorrect prioritisation of interventions	Medium	High	High	Mitigate	Establish evaluation criteria; compare appropriate metrics; validate model performance	AI/ML Lead	Open
R-04	Generative AI produces inappropriate or inaccurate responses	Ambiguous requirements, inadequate knowledge sources or weak controls	Patient misinformation, dissatisfaction or loss of trust	Medium	High	High	Mitigate	Define approved knowledge sources, response boundaries, guardrails and human escalation	GenAI Lead	Open
R-05	Patient intervention is ineffective	Intervention does not address actual attendance barriers	Limited improvement in appointment attendance	Medium	High	High	Mitigate	Base intervention design on analytical findings; test and refine intervention logic	Patient Engagement Lead	Open

ID	Risk	Cause / Trigger	Potential Impact	Likelihood	Impact	Rating	Response Strategy	Response / Mitigation	Owner	Status
R-06	Cross-workstream dependency failure	Delayed, incompatible or incomplete outputs between teams	Schedule slippage and rework	High	High	Critical	Mitigate	Maintain dependency register; establish integration checkpoints and escalation rules	Project Manager	Open
R-07	Scope expansion	Additional features or use cases introduced during development	Resource pressure, schedule delay and diluted objectives	Medium	High	High	Control	Apply formal change assessment against scope, schedule, resources and value	Project Manager	Open
R-08	Lack of baseline performance data	No validated pre-intervention metrics	Difficulty demonstrating project impact	High	Medium	High	Mitigate	Define baseline requirements and measurement methodology before intervention evaluation	Project Manager + Analytics	Open
R-09	Low stakeholder adoption	Limited participation, resistance or insufficient understanding	Reduced usability and implementation value	Medium	High	High	Mitigate	Engage stakeholders throughout requirements, design and validation	Project Manager	Open
R-10	Privacy or responsible-AI requirements are not adequately addressed	Inappropriate data handling or insufficient governance controls	Compliance, ethical and reputational exposure	Low–Medium	High	High	Avoid / Mitigate	Establish data-use boundaries, oversight requirements and AI governance controls	Responsible AI / Data Protection	Open
R-11	Workflow integration is technically or operationally impractical	Solution does not align with existing processes or system constraints	Functional solution fails to translate into operational use	Medium	High	High	Mitigate	Assess workflow fit before implementation; identify required process changes	Project Manager + Technical Leads	Open
R-12	Technical performance is prioritised over business value	Focus shifts toward model or system metrics alone	Technically successful project with limited operational benefit	Medium	High	High	Control	Evaluate technical, patient and operational outcomes together	Project Manager	Open
R-13	Limited project resources or time	Competing priorities or constrained team capacity	Delayed deliverables and reduced quality	Medium	Medium	Medium	Mitigate	Prioritise critical-path work; monitor workload and resource requirements	Project Manager	Open
R-14	Validation evidence is insufficient	Limited testing scenarios, user feedback or representative data	Uncertain solution readiness	Medium	High	High	Mitigate	Define validation criteria early and conduct structured testing before acceptance	Testing & Validation Lead	Open

2. RISK SCORING METHOD

Risk rating will be determined using:

$$\text{Likelihood} \times \text{Impact} = \text{Risk Rating}$$

Score	Rating	Management Response
1–4	Low	Monitor
5–9	Medium	Active mitigation
10–16	High	Priority management
17–25	Critical	Immediate management and escalation

Scoring Scale

Score	Likelihood / Impact
1	Very Low
2	Low
3	Moderate
4	High
5	Very High

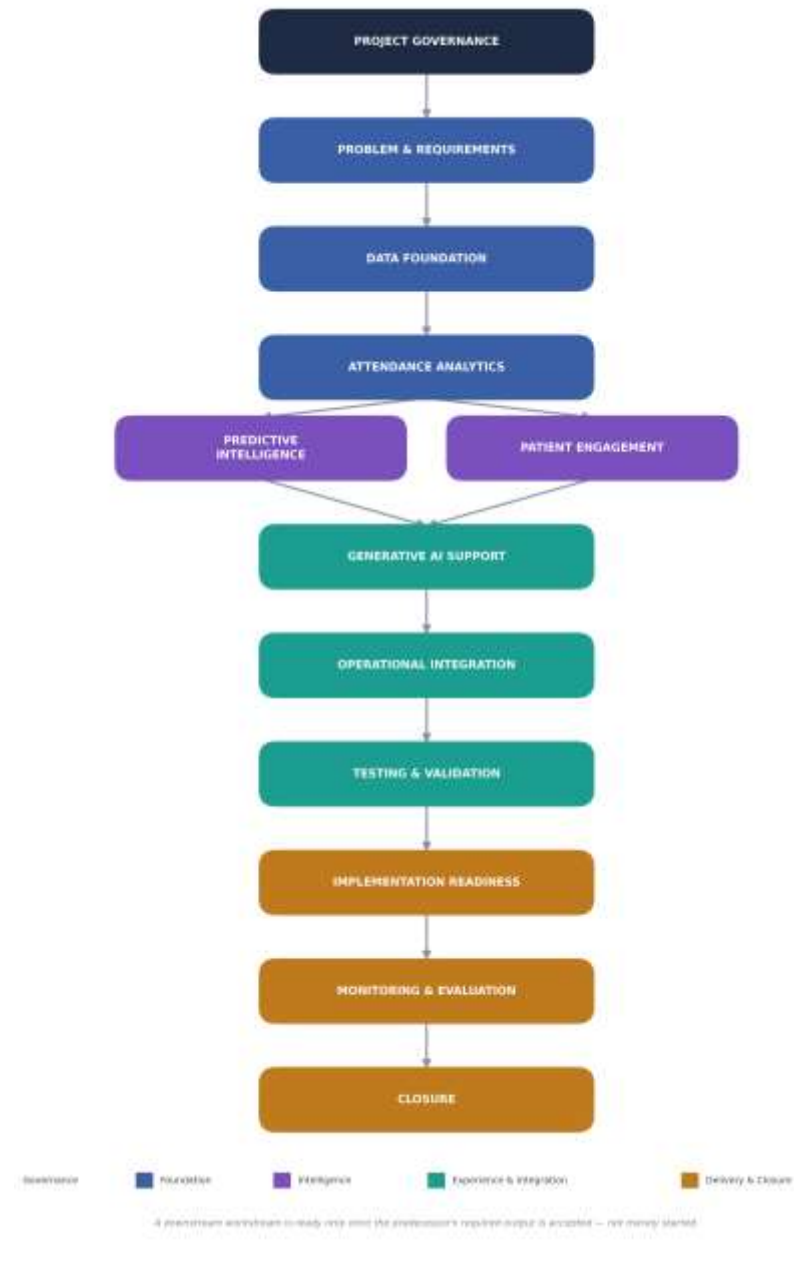
Risk scores will be reviewed as evidence becomes available.

3. KEY DEPENDENCY REGISTER

ID	Dependency	Predecessor	Dependent Activity / Deliverable	Type	Potential Impact if Delayed	Owner
D-01	Project scope and governance approval	Project Initiation	Problem & Requirements	<i>Finish-to-Start</i>	Requirements work cannot be formally baselined	Project Manager
D-02	Requirements definition	Problem & Requirements	Data Foundation	<i>Finish-to-Start</i>	Data collection and preparation may target the wrong requirements	Project Manager + Analytics
D-03	Appointment data availability	Data Access	Data Foundation	<i>Finish-to-Start</i>	Analysis cannot commence	Data & Analytics
D-04	Data quality assessment	Data Foundation	Attendance Analytics	<i>Finish-to-Start</i>	Analytical findings may be unreliable	Data & Analytics
D-05	Validated attendance insights	Attendance Analytics	Predictive Intelligence	<i>Finish-to-Start</i>	Model development may lack evidence-based features	AI/ML
D-06	Attendance insights	Attendance Analytics	Patient Engagement	<i>Finish-to-Start</i>	Intervention design may rely on assumptions	Patient Engagement
D-07	Approved patient-support requirements	Problem & Requirements	Generative AI Support	<i>Finish-to-Start</i>	GenAI use cases may be poorly defined	GenAI
D-08	Approved knowledge sources	Requirements / Information Review	Generative AI Support	<i>Finish-to-Start</i>	AI responses may lack reliable grounding	GenAI + Responsible AI
D-09	Predictive capability	Predictive Intelligence	Intervention Design	<i>Finish-to-Start</i>	Proactive intervention logic cannot be fully defined	AI/ML + Patient Engagement
D-10	Engagement and AI support components	Patient Engagement + GenAI	Operational Integration	<i>Finish-to-Start</i>	Integrated workflow cannot be completed	Project Manager
D-11	Defined future-state workflow	Operational Integration	Testing & Validation	<i>Finish-to-Start</i>	Testing cannot assess complete operational behaviour	Project Manager + Testing
D-12	Validation results	Testing & Validation	Implementation Readiness	<i>Finish-to-Start</i>	Implementation decision cannot be supported	Testing + Project Management
D-13	Defined outcomes and baseline measures	Analytics + Project Management	Monitoring & Evaluation	<i>Finish-to-Start</i>	Project impact cannot be reliably assessed	Project Manager + Analytics
D-14	Implementation readiness assessment	Implementation Readiness	Final Recommendations	<i>Finish-to-Start</i>	Recommendations may not reflect actual implementation conditions	Project Manager

4. CROSS-WORKSTREAM DEPENDENCY MAP

The principal dependency structure is shown below.



Critical Dependency Principle

A downstream workstream should not be treated as ready merely because its predecessor has started. Readiness should be based on the acceptance of the predecessor's required output. For example:

- Data Foundation Started ≠ Data Foundation Ready
- Model Developed ≠ Model Ready
- AI Capability Built ≠ AI Ready
- Integration Completed ≠ Solution Accepted

This distinction is particularly important for project control.

5. RISK & DEPENDENCY CONTROL

The Project Management function will maintain the register as a live control document throughout delivery.

Changes will be triggered by:

- New risks identified
- Existing risk probability or impact changing
- Risk mitigation becoming ineffective
- Dependency dates changing
- Dependency failure or blockage
- New cross-workstream dependencies emerging
- Scope or requirement changes
- Major technical or operational findings

Escalation Logic

Identify → Assess → Assign → Mitigate → Monitor → Escalate → Close

A risk or dependency will be escalated where its potential effect exceeds the responsible workstream's ability or authority to resolve it.

6. INITIAL RISK POSITION

At project initiation, the most significant exposures are:

1. Data Readiness

The availability and quality of appointment data directly affect the analytical and predictive workstreams.

2. Cross Work-stream Coordination

The project contains tightly coupled technical and operational outputs. Delays or changes in one workstream can propagate downstream.

3. Operational Fit

A technically functional solution may still fail if it does not align with actual clinic processes.

4. Responsible AI & Data Use

Patient-support applications require clear boundaries around information use, AI behaviour and human escalation.

5. Outcome Measurement

Without credible baseline measures, improvement cannot be demonstrated reliably.

These risks should receive priority during early project planning.

7. REGISTER CONTROL

Risk Register Owner	Project Management Function
Dependency Register Owner	Project Management Function
Review Frequency	At project control reviews and major stage gates
Escalation Trigger	Material effect on scope, schedule, resources, quality, risk or value
Status Values	Open / Monitoring / Escalated / Mitigated / Closed
Baseline Version	1.0
Next Review	Following initial requirements and data assessment

End of Initial Risk & Dependency Register